Boundary-Aware Network With Two-Stage Partial Decoders for Salient Object Detection in Remote Sensing Images

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***Abstract*— Salient object detection (SOD) is a binary pixelwise classification to distinguish objects in an image and also has attracted many research interests in the optical remote sensing images (RSIs). The existing state-of-the-art method exploits the full encoder–decoder architecture to predict salient objects in the optical RSIs, suffering from the problem of unsmooth edges and incomplete structures. To address these problems, in this article, we propose a boundary-aware network (BANet) with two-stage partial decoders sharing the same encoders for SOD in RSIs. Specifically, a boundary-aware partial decoder (BAD) is introduced at the first stage to focus on learning clear edges of salient objects. To solve the pixel-imbalance problem between boundary and background, an edge-aware loss is proposed to guide learning the BAD network. The resulting features are then used in turn to enhance high-level features. Afterward, the structure-aware partial decoder (SAD) is further introduced at the second stage to improve the structure integrity of salient objects. To alleviate the problem of incomplete structures, the structural-similarity loss is further proposed to supervise learning the SAD network. In a consequence, our proposed BANet can predict salient objects with clear edges and complete structure, while reducing model parameters due to the discardment of low-level features. Besides, training a deep neural network requires a large amount of images, and the current benchmark datasets for optical RSIs are not large enough. Therefore, we also create a large-scale challenging dataset for SOD in RSIs. Extensive experiments demonstrate that our proposed BANet outperforms previous RSI SOD models on all the existing benchmark datasets and our new presented dataset available at https://github.com/QingpingZheng/RSISOD.**

***Index Terms*— Boundary-aware network (BANet) with two-stage partial decoder, optical remote sensing image (RSI), salient object detection (SOD).**

I. Introduction

SALIENT object detection (SOD) targets at distinguishing the foreground and background regions from an image, having been of broad applications in many downstreaming visual tasks, such as image/video segmentation [1], [2], [3], video summarization [4], and image retargeting [5]. Recent decades have proved the remarkable advance of saliency detection in natural scene images [6], [7], but a limited number of researchers are exploring such a field in optical remote sensing images (RSIs). Compared with the natural scene images shot from a manually adjusted and close-range distance by hand-held cameras, optical RSIs are photographed outdoors in a high-altitude and high-angle shot via the satellite and aerial sensors. This results in an essential difference between salient object detection (SOD) in natural and remote sensing scenes. For example, RSIs cover more diverse scene patterns, more complicated backgrounds, and more diverse noise interference than natural scene images. Besides, salient objects in optical RSIs have the larger scale and orientation variations than those in the wild. Hence, directly transplanting the existing SOD models (trained on the natural image scene) into optical RSIs cannot achieve the optimal performance.

Similar to the natural image scene, the high quality of SOD in RSIs also requires having a smooth boundary and a complete spatial structure. However, the hardest challenge to SOD in RSIs is the collected image inevitably suffers from scale variability and low spatial resolution. To make salient objects adapt diverse scale variations, Li et al. [8] first proposed the LV-Net for SOD in optical RSIs, which uses an L-shaped encoder to capture the local details and group the complementary features and designs a V-shaped decoder for learning the distinct features automatically. Nevertheless, such a model is still not accurate enough to locate salient objects, because the complex background (e.g., objects with imaging shadows) often interferes and corrupts the structural integrity of salient objects. In addition, when the spatial topology of the salient objects is relatively complex, the existing LV-Net fails to detect salient objects completely since the convolution operation is incapable of capturing the long-range semantic dependencies between pixels [9]. To alleviate these issues, Zhang et al. [10] proposed the DAFNet model, which devises a dense attention fluid (DAF) structure to explicitly connect the attentive features at different levels, exploits the low-level attention cues to generate the high-level attentive features, and integrates the point-to-point semantic relationships into the global-context information to align the features at distant spatial positions.

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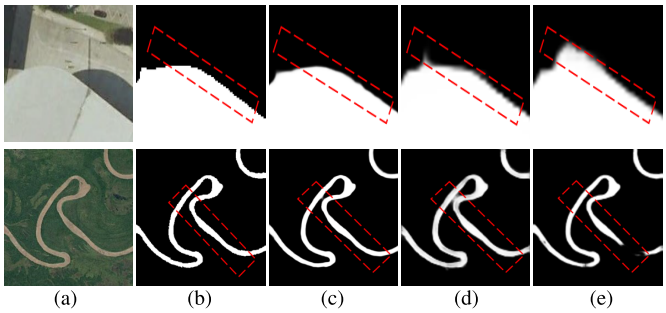

Fig. 1. Visual comparison of saliency maps generated by our proposed BANet, DAFNet, and LV-Net, demonstrating the problems brought from DAFNet and LV-Net. The first row shows that the salient objects generated by DAFNet and LV-Net have unsmooth edges. The second row shows that the salient objects generated by DAFNet and LV-Net suffer from incomplete spatial structure (also dubbed aperture problem). (a) Image. (b) GT. (c) BANet. (d) DAFNet. (e) LV-Net.

Although the existing DAFNet achieves satisfactory performance on SOD in optical RSIs, it still has two drawbacks of the rough surface and incomplete spatial structure (aperture problem). The visual examples in Fig. 1 present the saliency maps predicted with respect to LV-Net and DAFNet. The first and second rows of Fig. 1 show the resulting salient objects with uneven edges and a severe aperture problem, respectively. This is due to the fact that the current SOD models [8], [10] in RSIs are based on the full encoder–decoder convolutional neural networks [11] which integrate all the low-level features with high-level features at the decoder stage [12]. Nonetheless, introducing too many low-level features in the decoder network contributes less to the final performance while expending a large computational overhead.

Inspired by the cascaded partial decoder (CPD) framework [12] which solely uses the partial deep features to recover spatial details of salient objects, we propose a boundary-aware network (BANet) with two-stage partial decoders for SOD in optical RSIs to alleviate the above-mentioned issues. In particular, our proposed BANet comprises a partial boundary-aware decoder (BAD), a boundary-guided feature enhancement (BGFE) module, and a structure-aware partial decoder (SAD). To obtain the salient objects with clear boundary, we introduce the edge-aware binary dice loss to guide learning of the BAD branch, solving the severe pixel-imbalance problem between the foreground and the background in the optical RSIs. The resulting features are then adopted in turn to refine the first level of high-level features, since the use of downsampling operation in the BAD network can probably filter out the boundary of small salient objects. To further obtain the salient objects with complete structure, we design another structure-aware loss function to supervise learning the SAD branch. Consequently, our proposed BANet can generate high-quality salient objects with clear boundary and complete structure, while reducing the computational cost.

Besides, training a deep-learning-based network requires massive high-quality data to prevent overfitting. Three SOD benchmark datasets have been developed specially for SOD model training and performance evaluation in optical RSIs, including ORSSD [8], EORSSD [10], and ORSI-4199 [13]. The ORSSD dataset is challenging, due to diverse spatial resolution, cluttered background, and various types of salient objects, but it contains only 800 images in total and is relatively small. The EORSSD dataset extends the ORSSD to a slightly larger one (including 2000 images split into 1200 training and 800 testing images) by adding more varied object attributes, more complicated types of scenes, and more comprehensive circumstances in the real world. However, it is still not large enough to train a deep-learning-based network. To address these problems, a large-scale ORSI-4199 dataset (with 4199 images) is created by collecting the images from the same source as the ORSSD and the EORSSD. Nonetheless, it covers limited scenes. In this article, we construct a new large-scale dataset which comprises the optical RSIs for SOD (dubbed RSISOD), including around 5 K images and covering more challenging scenes, such as multiple different object attributes simultaneously recognized as salient objects in an image, more diverse small-scale objects, and more cluttered background.

The main contributions can be summarized as follows.

1) An end-to-end BANet with two-stage partial decoders is proposed for SOD in optical RSIs, including one partial decoder for learning the boundary of salient objects, a BGFE module, and another partial decoder for predicting the final saliency map with complete spatial structures.
2) An edge-aware dice loss is proposed to supervise learning the boundary-aware partial decoder by considering the pixel imbalance between salient and non-salient objects, while the structural-similarity loss is introduced to supervise the learning of the final saliency map with complete spatial topology as much as possible.
3) A new challenging and large-scale dataset is constructed for SOD in the optical RSIs, including around 5 K images with the corresponding pixelwise annotations.

The rest of this article is organized as follows. Section II briefly reviews the related works on SOD in both NSIs and optical RSIs. Sections III and IV elaborate on the details of the proposed BANet and the new RSISOD benchmark dataset, respectively. Section V discusses the experimental comparisons and ablation studies, and finally Section VI concludes this article.

II. Related Work

In this section, we first review the traditional SOD methods for optical RSIs (RSIs) and then present the existing convolutional neural network (CNN)-based SOD models for RSIs.

A. Traditional SOD Methods for RSIs

Similar to the conventional SOD techniques in the natural image scenes, the traditional SOD methods [14], [15], [16], [17], [18] in RSIs mainly rely on the bottom-up stimulus-driven approaches which explore hand-crafted features based on various saliency assumptions. For example, Faur et al. [19] reported a rate distortion estimation method which considers the men-shift algorithm for RSI segmentation. Zhao et al. [20] achieved salient object prediction

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based on the use of high-level global and background priors. Zhang et al. [18] introduced the color prior into RSIs, calculated scores of saliency map on each color channel, and integrated the color components to achieve saliency detection in RSIs. Boa Zhang et al. [16] represented a self-adaptive fusion method based on low-rank matrix recovery, which fuses color information, intensity information, texture information, and global contrast information for final results. Huang et al. [15] developed a VHR RSI saliency detection method based on a contrast-weighted dictionary learning.

However, such hand-crafted feature-based methods have the limitations on a large amount of computational cost and memory consumption. Besides, it is difficult to design the effective hand-made features based on human priors. Therefore, the fore-mentioned traditional SOD approaches in RSIs are not efficient enough for practical applications.

B. CNN-Based SOD Models for RSIs

With the emergence of deep-learning-based methods [21], [22], [23] in the optical RSIs, CNN-based models [10], [24], [25], [26] have also become a mainstream framework for SOD in the optical RSIs, especially for the fully convolutional network (FCN)-based encoder–decoder architectures. For instance, Li et al. [8] proposed an end-to-end FCN-based encoder–decoder architecture with a two-stream pyramid module to achieve the first deep-learning-based SOD model for the optical RSIs. Zhang et al. [10] introduced the DAFNet model which leverages a DAF structure to explicitly connect different-level features to generate high-level attentive features. Li et al. [24] proposed a coarse-to-fine CorrNet model which exploits high-level semantic features to obtain the initial course saliency map and then uses low-level detailed features to refine it for final fine result. Li et al. [25] designed an adjacent context coordination-based encoder–decoder architecture for saliency prediction in RSIs. Zhou et al. [27] developed an edge-aware recurrent positioning network to detect salient objects in the optical RSIs.

All the above models progressively aggregate all the low-level features into the final high-level saliency maps, achieving a relatively surprising performance, but consuming too much time for saliency prediction. Since low-level features contain more spatial details and high-level features provide more semantic information, some researchers argue that high-level semantic features always contribute more than low-level features for a dense prediction task, while fusing more low-level features into final results needs more computation cost. Based on this argumentation, Zhe et al. [12] proposed a CPD framework without consideration of low-level features, for the fast and accurate SOD in natural image scenes. Inspired by the CPD model, this article introduces the two-stage partial decoders for SOD in optical RSIs.

III. Proposed Method

This section depicts the proposed BANet. Section III-A presents the network overview of our BANet, which comprises a partial BAD, a BGFE module, and an SAD. Sections III-B–III-D elaborate the details of the BAD, the BGFE module, and the SAD, respectively.

A. Overall Framework

The objective of the SOD dense prediction task requires predicting salient objects with clear edges and complete spatial topologies. To achieve this goal, most existing deep-learning-based SOD models for the optical RSIs are developed based on the conventional encoder–decoder architecture, which aggregates all the encoded low-level and high-level features into the decoder modules in a hierarchical manner. However, these models cost much computation overhead. To alleviate this problem, we propose a BANet, which only aggregates the high-level features into decoders. Compared with the traditional encoder–decoder networks, the proposed BANet can reduce the computational complexity because of dropping the low-level features.

Fig. 2 describes the overall framework of our proposed BANet (BANet), which is a two-stage partial encoder–decoder architecture for SOD in RSIs. Given an $H \times W$ RSI image, it flows into the VGG16 [28] backbone network and is encoded into multilevel features $\{f_1, f_2, f_3, f_4, f_5\}$ with scales corresponding to $H \times W$, $(H/2) \times (W/2)$, $(H/4) \times (W/4)$, $(H/8) \times (W/8)$, and $(H/16) \times (W/16)$, respectively. Of them, the shallower features $\{f_1, f_2\}$ contain much details but contribute less to the final performance. Moreover, low-level features have a large resolution, leading to a high computational cost. Compared with low-level features, high-level features $\{f_3, f_4, f_5\}$ contribute more information to the final result for the dense prediction task.

To obtain the high-performance SOD model with low computational cost, we propose an end-to-end two-stage partial decoders, including one partial BAD and another SAD. Both the decoders share the same low-level features $\{f_1, f_2\}$. Particularly, the first partial decoder BAD focuses on the generation of an edge-aware saliency map, and it is learned by our proposed pixel-imbalanced surface loss. The resulting map is then used in turn to enhance the high-level semantic features f_3, alleviating the pixel-imbalanced problem of these feature maps. At the second stage, the enhanced high-level features flow into the SAD branch to obtain the final saliency map. To make it have a complete structure as much as possible, a structural-similarity loss is further introduced to learn the partial decoder of SAD. Consequently, two decoders are trained simultaneously, and the total loss function of our BANet is defined as

$$\mathcal{L}_{\text{BANet}} = \mathcal{L}_{\text{BAD}} + \mathcal{L}_{\text{SAD}} \tag{1}$$

where $\mathcal{L}_{\text{BAD}}$ and $\mathcal{L}_{\text{SAD}}$ denote the loss function for the BAD branch and the SAD branch, respectively.

B. Boundary-Aware Partial Decoder

Compared with the conventional encoder–decoder architecture, the BAD branch only uses the high-level features $\{f_3, f_4, f_5\}$ as inputs and adopts a progressive way to restore the scales and spatial topology of these features. To enlarge receptive field, convolution kernels with different sizes are used to convolve the same features, because the convolution operation can only capture limited contextual information with a fixed kernel size. Specifically, each of the high-level

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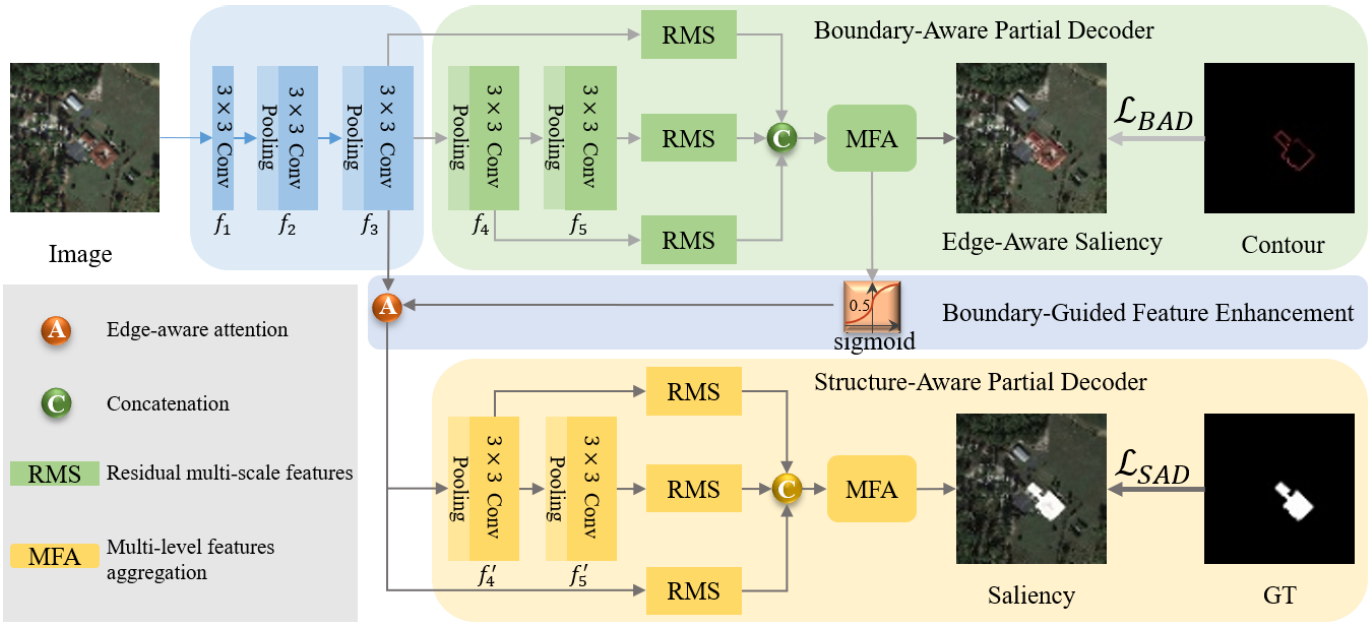

Fig. 2. Overview framework for SOD in RSIs. The proposed BANet is a BANet with two-stream partial decoders, including (a) BAD which focuses on the generation of an edge-aware saliency map supervised by the balanced edge-aware loss $\mathcal{L}_{\text{BAD}}$, (b) BGFE module which leverages the boundary-aware saliency map to achieve an edge-aware attention, and (c) SAD which is supervised by the balanced structural-similarity loss $\mathcal{L}_{\text{SAD}}$ and predicts the final saliency map with complete spatial structures.

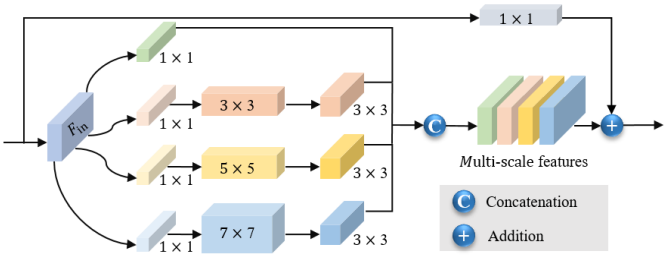

Fig. 3. RMS module, using 1 × 1, 3 × 3, 5 × 5, and 7 × 7 kernel sizes to generate multiscale features.

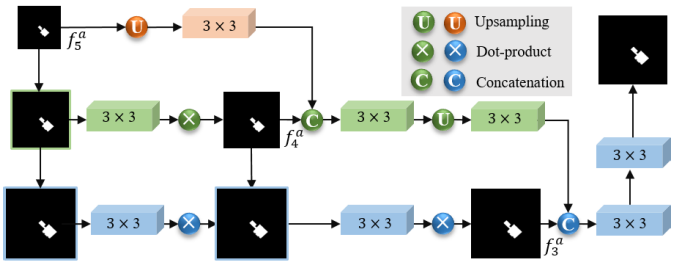

Fig. 4. MFA module is used to fuse different levels of features after preprocessing by the RMS modules.

features $\{f_3, f_4, f_5\}$ is prepossessed by the proposed residual multiscale (RMS) module shown in Fig. 3, resulting in a new set of multiscale features $\{f_3^a, f_4^a, f_5^a\}$ with spatial resolutions of $(H/4) \times (W/4)$, $(H/8) \times (W/8)$, and $(H/16) \times (W/16)$, respectively.

Afterward, a hierarchical and progressive strategy is introduced to aggregate these multiscale and multilevel features step by step at the stage of the partial decoder. Fig. 4 elaborates the whole process of our developed multilevel feature aggregation (MFA). Initially, the high-level features f_5^a are upsampled to the same spatial size as the features f_4^a and the upsampled features are further passed into a 3 × 3 convolution. The resulting features are then merged with features f_4^a via a dot product, leading to a new feature f_4^b. Successively, we continue to upsample and convolve features f_5^a again using the same operations, concatenate the resulting features with previous features f_4^b, and manipulate them by another 3 × 3 convolution, generating the features f_4^c. Similarly, features f_5^a and f_4^a are upsampled to the same spatial size of features f_3^a, respectively, and the resulting features are input into the corresponding convolution with 3 × 3 kernel size. The resulting features are then combined with features f_3^a to obtain new features f_3^b by the dot-product operation. The obtained features f_4^c are upsampled two times and convolved by a 3 × 3 convolution, and the resulting features are concatenated with features f_3^b. These concatenated features successively flow into two 3 × 3 convolution and one 1 × 1 convolution, obtaining an aggregated saliency map M_{BAD}.

Compared with the salient objects in NSIs, salient objects in RSIs are much smaller, resulting in a much more severe problem of pixel imbalance between the foreground and background information. To address this issue, we introduce a binary dice loss, which can balance the pixels between the foreground and the background. Meanwhile, it can prevent the training process from getting trapped in local minima, without the need for human-determined sample reweighting [29]. Herein, the binary dice loss is defined as

$$\mathcal{L}_{\text{dice}} = \frac{1}{N}\sum_{i=1}^{N}\left[1 - \frac{\sum g \cdot p}{\sum\left(g^2 + p^2\right)}\right] \tag{2}$$

where g and p correspond to the ground truth and predicted saliency map, respectively, and N represents the total amount

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of testing samples. Furthermore, to make this decoder branch focus on learning the boundary of salient objects as clearly as possible, we introduce another surface loss (or edge-aware loss), defined as

$$\mathcal{L}_{\text{edge-aware}} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{e_i} * \mathcal{L}_i^e \odot E_i \tag{3}$$

where e_i is the number of boundary pixels in an edge label map E_i, and $\mathcal{L}_i^e$ is the cross-entropy between a predicted saliency map and the ground-truth. Consequently, the total edge-aware binary dice loss is computed as

$$\mathcal{L}_{\text{BAD}} = \mathcal{L}_{\text{dice}} + \mathcal{L}_{\text{edge-aware}}. \tag{4}$$

C. Boundary-Guided Feature Enhancement

Inevitably, the saliency map originated from the partial decoder of BAD network probably contains some distractors, suppressing the prediction accuracy on salient objects, especially when the RSIs have a relatively complex background. Although the BAD branch can well preserve the edges of salient objects, the boundary for small salient objects in optical RSIs is more likely to be filtered out due to the use of deep downsampling operations. To solve these problems, this article puts forward a BGFE module, which adopts the resulting boundary-aware saliency map as an attentive cue to refine the original high-level features f_3 in turn, achieving an edge-aware attention mechanism.

Concretely, the edge-aware saliency map is first processed by a Gaussian convolution Conv_g, which is a convolution operation with a Gaussian kernel and zero bias. The resulting map is then normalized into the range of [0, 1] via a sigmoid function, obtaining the normalized attention map. Eventually, the maximum element of the attention map is multiplied by the original features f_3, filtering many useless distractors out of the features f_3. To get more spatial information, the enhanced features can be further added with themselves, assigning different weights to salient and non-salient pixels and standing out the salient region. The overall procedure of the BGFE module can be defined by

$$f_3^h = \text{Max}\{\text{Norm}[\text{Conv}_g(M_{\text{BAD}})]\} \cdot f_3 + f_3 \tag{5}$$

where M_{BAD} denotes the edge-aware saliency map from the BAD network.

D. Structure-Aware Partial Decoder

The saliency map originated from the BAD branch is probably unable to recover its spatial topology completely, and the salient regions within the enhanced high-level features f_3^h are still too coarse. Therefore, we use another SAD to refine the features f_3^h and generate a final saliency map with a smooth edge and complete spatial structure as much as possible. The whole decoder process is almost the same as the BAD, but the only difference is that the aggregating features used in this process are $\{f_3^h, f_4^h, f_5^h\}$ instead of $\{f_3, f_4, f_5\}$.

Different from the BAD, another partial decoder of SAD mainly aims to restore the complete spatial structure of the salient objects as much as possible, without using any low-level detail. To achieve this objective, we introduce a class-balanced structural-similarity loss to supervise the SAD branch just learning on the spatial topology. The reason for adopting the class-balanced loss herein is that at the processing of the BGFE module, the enhanced features f_3^h have already eliminated the problem of pixel imbalance between the salient and non-salient regions using a self-addition operation for automatically allocating different weights to each salient or non-salient pixel.

Here, we use the SSIM [30] loss originally used for image quality assessment to capture the structural topology of salient objects in optical RSIs. Suppose that the predicted saliency map is s and the corresponding ground truth is g, the SSIM loss function can be written as

$$\mathcal{L}_{\text{ssim}} = 1 - \frac{(2\mu_s\mu_g + C_1)(2\sigma_{sg} + C_2)}{(\mu_s^2 + \mu_g^2 + C_1)(\sigma_s^2 + \sigma_g^2 + C_2)} \tag{6}$$

where μ_s, μ_g and σ_s, σ_g correspond to the mean and standard deviations of s and g, respectively, σ_{sg} denotes the covariance between them, and constants C_1 and C_2 are, respectively, set to 0.01^2 and 0.03^2, avoiding dividing by zero.

In addition, we further use the class-balanced BCE [31] loss widely used for binary classification to achieve pixelwise segmentation. Following the above assumption, the BCE loss function can be expressed as

$$\mathcal{L}_{\text{bce}} = -\sum\big[g \cdot \log(s) + (1-g)\log(1-s)\big]. \tag{7}$$

Then, the whole structure-aware loss function for N testing samples can be computed as

$$\mathcal{L}_{\text{SAD}} = \frac{1}{N}\sum_{i=1}^{N}\big(\mathcal{L}_{\text{bce}}^i + \mathcal{L}_{\text{ssim}}^i\big) \tag{8}$$

where $\mathcal{L}_{\text{bce}}^i$ is the BCE loss of the ith sample, and $\mathcal{L}_{\text{ssim}}^i$ represents the SSIM loss of the ith sample.

IV. Proposed Dataset

This section constructs the new optical RSIs for SOD (RSISOD). Section IV-A presents the comparison analysis of the existing benchmark datasets for SOD in the optical RSIs and elaborates the reason to propose our RSISOD dataset for optical SOD. Section IV-B provides several visual examples and statistical analysis to demonstrate the challenge and advantage of our newly created RSISOD dataset.

A. Dataset Comparison for Optical RSIs

Until now, there are only three public datasets available for SOD in RSIs. ORSSD [8] is the first public dataset built for optical remote sensing saliency detection, containing only 600 images for training and 200 images for testing. Since it is insufficient to train a deep-learning-based model, EORSSD [10] extends it to 2000 images, having 1400 training images and 600 testing images. Besides, the EORSSD dataset is more challenging than ORSSD by adding images with more salient objects, images with small salient objects, images with abundant scenarios, and so on. However, the amount

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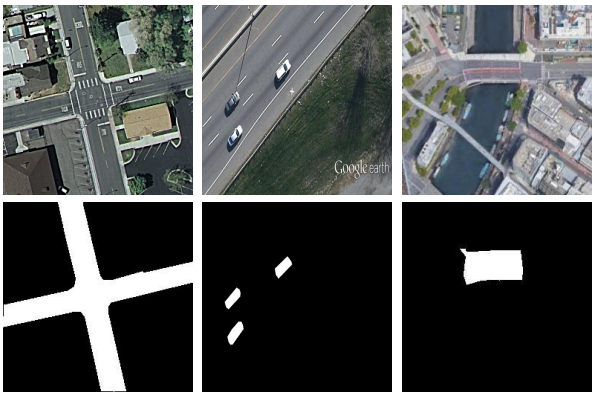
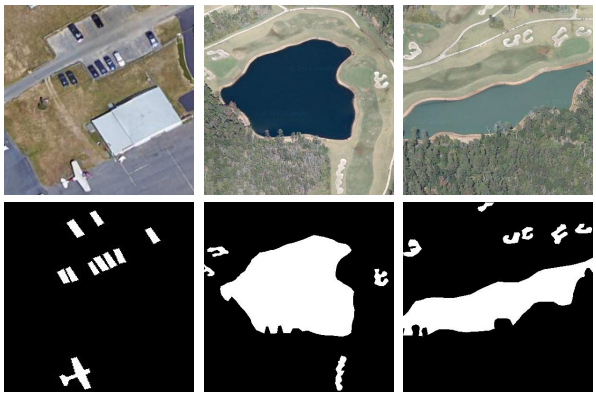


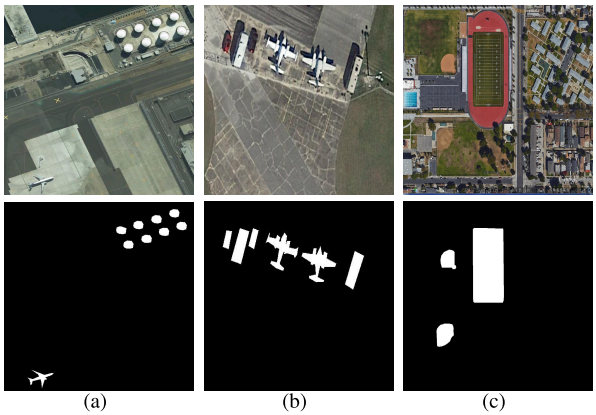
(a) (b) (c)

Fig. 5. Examples of samples with different types of salient objects. (a)–(c) Just list an example of new created dataset.



(a) (b) (c)

Fig. 6. Examples of samples with dense salient objects in the RSISOD dataset. (a)–(c) Just list an example of new created dataset.



(a) (b) (c)

Fig. 7. Examples of samples with more than one salient object in a scenario. (a)–(c) Just list an example of new created dataset.

of data is still slightly too small to train a deep learning model. To solve this problem, ORSI-4199 [13] introduces more images. Nonetheless, it covers limited scenes.

To enlarge the data size and raise the challenge for the SOD task in the optical RSIs, we construct a new large-scale RSISOD dataset with almost 5000 images and the corresponding pixelwise annotation for each image. Compared with the existing benchmark datasets, it covers more real-world scenes, more various object attributes, and more types of salient objects. To construct such a dataset, we collect the original optical RSIs from several existing RSI databases, such as LEVIR dataset [32], Google Earth, and NWPU datasets [33], [34]. Post hoc, we use a general annotation tool (labelme) to manually label the pixelwise saliency mask for each image. For uniform sampling, we randomly pick up 75% images from each scene for training, and the remaining images for testing. Eventually, the whole dataset is divided into two parts: 3784 images for training and 1270 images for testing.

B. Dataset Statistics and Analysis

1) Challenge in More Types of Salient Objects: In the RSISOD dataset, we enrich the types of salient objects, such as bridges, crossing roads, and helipads, increasing the difficulty of model generalization. The main type of salient objects includes 17 types of attributes, namely, baseball, beach, bridge, building, car, crossroad, golf, helipad, house, oil tank, plane, playground, swimming pool, river, ship, stadium, and wastewater. Moreover, these attributes can interfere with each other. For example, as shown in Fig. 5(a) and (b), objects car and bridge appear in the same image, but only the bridge or the car is labeled as the salient object.

2) Challenge in More Dense Salient Objects: In the RSISOD dataset, we add a type of data in that an image contains more salient objects than the previous two datasets. Compared with the ORSSD and EORSSD datasets, the RSISOD increases the difficulty of model prediction largely, due to smaller size of an object, more dense salient objects in an image, and more variety of ratios between the foreground and the background. For example, the salient objects cars in Fig. 6(a) are dense and small. As shown in Fig. 6(b) and (c), the ratio between the river and the non-salient region changes by a large margin.

3) Challenge in More Than One Type of Salient Objects in a Scenario: Different from other optical SOD datasets where one image normally contains only one type of salient object, our constructed RSISOD dataset provides an extra class of case where more than one type of object is taken into account as salient objects in an image, challenging the model predicting ability for such a complex scenario. For example, as shown in Fig. 7(a)–(c), each image contains two different types of salient objects. Furthermore, each type of salient object is quite small or dense.

4) Challenge in Salient Objects With More Various Scales: To increase the challenging difficulty of modeling in recognizing salient objects of diverse scales, the RSISOD dataset supplies some data samples where the same type of salient objects has a very different scale variation. For example, as shown in Fig. 8(a)–(c), the shape of the river is changed largely, runs throughout the whole image, and has various scales. Fig. 8(b) provides a visual sample that different rivers have relatively different scales.

5) Challenge in Salient Objects With More Clustered Background: In the SOD task, salient objects suffer a different extent of disturbance from the background. The difficulty of the SOD increases as the non-salient region becomes complicated. For example, Fig. 9(a) shows that the non-salient region exits some object with a very similar shape to the salient object. Fig. 9(b) shows the salient object has a very

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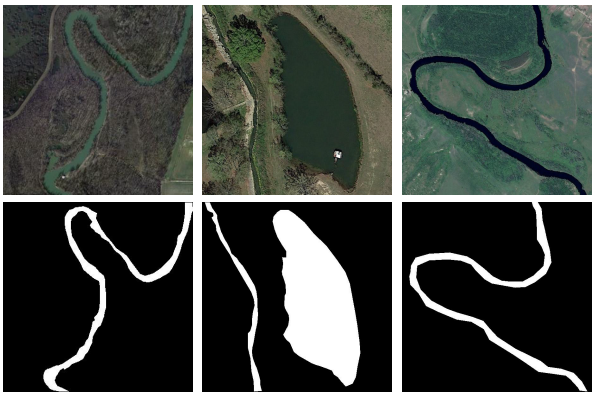
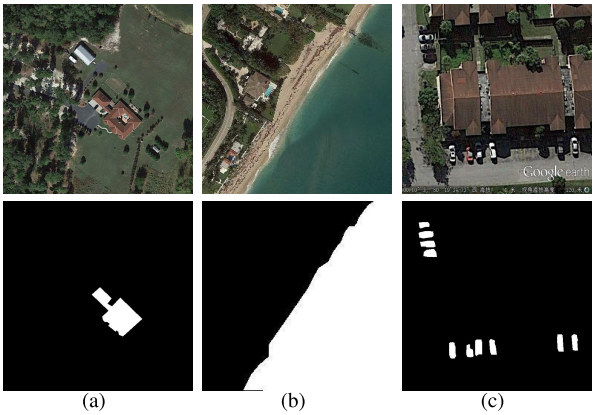


(a) (b) (c)

Fig. 8. Examples of samples with different scales of salient objects. (a)–(c) Just list an example of new created dataset.



(a) (b) (c)

Fig. 9. Examples of samples with different clustered background. (a)–(c) Just list an example of new created dataset.

similar context as the background. Fig. 9(c) provides an visual example that small salient objects have both the same shape and same context as the non-salient region.

In summary, our constructed RSISOD dataset is over twice larger than EORSSD, conducive to training the deep-learning-based SOD models. Meanwhile, it covers more comprehensive and challenging scenarios than the off-the-shelf datasets.

V. Experiments

A. Dataset

We evaluate the performance of the proposed BANet method on the ORSSD dataset [8], the EORSSD dataset [10], the ORSI-4199 dataset [13], and our constructed RSISOD dataset. ORSSD contains 600 images for training and 200 images for testing. EORSSD is an extension of ORSSD, including 1200 training images and 800 testing images. ORSI-4199 has 2000 images for training and 2199 images for testing. Our newly created RSISOD dataset contains 5054 images with the corresponding annotations, and we randomly selected 3784 images from the constructed RSISOD dataset for training and the rest of 1270 images as the testing dataset.

B. Evaluation Metrics

Similar to the SOD in natural salient images, this article also uses F-measure, MAE score, and S-measure to evaluate the optical SOD performance for quantitative comparisons of different methods. The F-measure and S-measure metrics are the bigger the better. The MAE metric is the smaller the better. To calculate the evaluation metric between the GT and the predicted saliency map, we have to convert the obtained saliency map into several binary masks using a set of thresholds (integers from 0 to 255).

The F-measure is a weighted harmonic of precision and recall, defined as

$$\mathrm{F}_\beta = \frac{\left(1+\beta^2\right)\mathrm{Precision} \times \mathrm{Recall}}{\beta^2 \mathrm{Precision} + \mathrm{Recall}} \tag{9}$$

where $\beta^2 = 0.3$ is assigned to allocate more weight to precision than recall. For each predicted saliency map, a group of F-measure values can be calculated by a series of thresholds ranging from 0 to 255, and the average F-measure score can be obtained by taking the mean of these values. Calculating all the average scores of the F-measure for all the predicted images corresponding to different thresholds, the maximum mean F-measure is selected as a final evaluation index. The larger the F-measure, the better performance the model achieves.

The MAE metric calculates the average difference between a final continuous saliency map $\overline{S}$ and a binary ground-truth $\overline{G}$, by considering both salient and non-salient pixels, and it is defined as

$$\mathrm{MAE} = \frac{1}{(w \times h)} \sum_{x=1}^{w} \sum_{y=1}^{h} \left\| \overline{S}(x, y) - \overline{G}(x, y) \right\| \tag{10}$$

where w and h denote the width and the height of an input frame, respectively, and both the saliency map $\overline{S}$ and ground truth $\overline{G}$ are normalized to the values between 0 and 1. The smaller the MAE score is, the better performance the model has.

The S-measure captures the similarity over nonbinary foreground maps, comprising a region-aware structural similarity and an object-aware structural similarity, which is defined by

$$S = \alpha \cdot S_o + (1-\alpha) \cdot S_r \tag{11}$$

where $\alpha \in [0, 1]$ and α is herein empirically set to 0.5. The further details of computing S_o and S_r are omitted here but can be found in [35]. The larger value of the S-measure means better performance.

C. Implementation Details

We implement the proposed BANet based on the PyTorch framework, running on a workstation equipped with an NVIDIA Tesla v100 GPU for acceleration. The network takes VGG16 [28] as a backbone and is trained using the ADAM optimization strategy for 100 epochs, and the batch size is set to 10. In the training phase, the learning rate is initialized as 10^{-4} and decreased by 10% when the training loss becomes flat. For a fair comparison, the same datasets are used to train our proposed model and other methods. The data augmentation techniques used herein are the same as DAFNet, involving combinations of flipping and rotation and producing seven variants for every single sample. All the training and testing images are uniformly resized to 352 × 352. Besides,

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TABLE I

QUANTITATIVE COMPARISONS WITH 13 STATE-OF-THE-ART DEEP SOD MODELS IN RSIS ON EORSSD, ORSSD, ORSI-4199, AND OUR PROPOSED RSISOD DATASETS. THE TOP THREE RESULTS ARE HIGHLIGHTED IN **RED**, **BLUE**, AND **GREEN**, RESPECTIVELY. ↑ AND ↓ DENOTE LARGER AND SMALLER ARE BETTER, RESPECTIVELY. NOTE "L.W" REPRESENTS LIGHTWEIGHT

Methods	Backbone	Speed	#Param	FLOPs	ORSSD [8]			EORSSD [10]			ORSI-4199 [13]			RSISOD		
		(FPS) ↑	(M)↓	(G)↓	$S_\alpha\uparrow$	$F_\beta^{\max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$F_\beta^{\max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$F_\beta^{\max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$F_\beta^{\max}\uparrow$	$\mathcal{M}\downarrow$
$LVNet_{19}$ [8]	VGG16	1.4	54.3	-	.8815	.8263	.0207	.8630	.7794	.0146	-	-	-	-	-	-
$DAFNet_{21}$ [10]	VGG16	51	29.4	137.02	.9191	.8928	.0125	.9167	.8614	.0060	.8537	.8308	.0427	.8798	.8807	.0284
$EMFINet_{21}$ [37]	VGG16	54	107.3	101.99	.9366	.9002	.0109	.9290	.8720	.0084	.8746	.8572	.0315	.9028	.8966	.0221
$MJRBM_{21}$ [13]	VGG16	57	43.5	158.46	.9204	.9014	.0141	.9197	.8765	.0095	-	.8628	.0373	-	-	-
$SARNet_{21}$ [38]	VGG16	54	25.9	236.29	.9134	.8850	.0187	.9240	.8719	.0099	.8525	.8328	.0424	.9015	.9050	.0220
$RRNet_{21}$ [39]	VGG16	51	-	-	.9274	.9186	.0118	.9195	.9016	.0078	.8637	.8450	.0385	.8984	.8998	.0233
$MCCNet_{21}$ [40]	VGG16	60	67.7	234.05	.9437	.9155	.0087	.9327	.8904	.0066	.8723	.8573	.0323	.9080	.9080	.0206
$CorrNet_{22}$ [24]	L.W VGG	78	4.1	42.16	.9380	.9129	.0098	.9289	.8778	.0083	.8529	.8363	.0427	.8870	.8756	.0257
$PGNet_{22}$ [41]	VGG16	34	43.5	95.72	.9255	.9194	.0112	.9255	.8885	.0044	.8845	-	.0321	-	-	-
$FSMINet_{22}$ [42]	-	-	3.6	23.58	.9361	.9041	.0101	.9255	.8678	.0079	.8611	.8381	.0378	.8968	.8929	.0219
$ACCoNet_{22}$ [25]	VGG16	56	102.6	368.59	.9437	.9149	.0088	.9290	.8837	.0074	.8751	.8549	.0335	.9075	.9069	.0213
$ERPNet_{22}$ [27]	VGG16	72	56.5	174.40	.9254	.8974	.0135	.9210	.8632	.0089	.8637	.8436	.0356	.9028	.8969	.0211
$MSCNet_{22}$ [43]	MobileNet	-	3.3	23.38	.9226	.9107	.0132	.9086	.8806	.0090	.8659	.8474	.0367	.9013	.8974	.0224
BANet(Ours)	VGG16	70	29.2	118.85	.9432	.9205	.0097	.9292	.9055	.0089	.8767	.8576	.0314	.9083	.9073	.0205

this article does not use any postprocessing procedure (e.g., CRF [36]).

D. Comparison With State-of-the-Art Methods

In this article, our BANet is compared with 13 state-of-the-art deep models for SOD on the testing subset of THE ORSSD dataset, EORSSD dataset, ORSI-4199 dataset, and RSISOD dataset, including LVNet [8], DAFNet [10], EMFINet [37], MJRBM [13], SARNet [38], RRNet [39], MCCNet [40], CorrNet [24], PGNet [41], FSMINet [42], ACCoNet [25], ERPNet [27], and MSCNet [43]. The predicted saliency maps of the ORSSD, EORSSD, and ORSI-4199 datasets are provided by the authors directly, and the results of our constructed RSISOD dataset are generated by the sources codes. The proposed method is trained using the same training dataset as all the other deep-learning-based methods.

1) Quantitative Comparison: Table I quantitatively shows comparison of the proposed method and the existing deep learning methods on the three widely used metrics. The results in Table I show that our proposed BANet has higher or comparative performance than previous state-of-the-art method deep models on all the benchmark datasets. Our BANet outperforms all the other methods on F-measure scores over 1.1‰ and 3.9‰ on ORSSD and EORSSD respectively. On ORSI-4199, our method ranks 1st on both F-measure and MAE scores. On the new constructed RSISOD dataset, the proposed method also achieves the best performance on both S-measure and MAE scores. For other metrics, our method almost ranks the 2nd and 3rd results. This is attributed to our proposed two-stage partial decoders, affirming that the high-level features contribute more effective information to the final results than the low-level features. Besides, our BANet has 29.2 M parameters and 11 885G flops and achieves around 70 frames/s for inference time. Compared with other methods using the same VGG16 backbone, BANet ranks the 2nd or 3rd model complexity, verifying the efficiency of our proposed framework.

2) Qualitative Comparison: Figs. 10 and 11 visually compare the qualitative results of our proposed BANet and the other state-of-the-art methods. We observe that our method has a superior ability to predict salient objects with more clear edges and more complete spatial structures than other methods. In Fig. 11, the first column images show that our method can easily separate two close objects. The second column images demonstrate that BANet generates salient objects with more clear edges and complete spatial typologies than others. In Fig. 10, the images lying in the third and fourth columns have better performance in the small and dense scenarios. The fifth and sixth column images show that our method is also superior to others in complicated objects. The last three columns present that the proposed BANet generates salient objects with the most smooth contours and the best structural integrity in complex scenarios.

E. Ablation Study

In this section, we conduct ablation experiments to validate the effectiveness of our critical components, including boundary-aware partial decoder (BAD), BGFE module, and SAD. The baseline model uses VGG16 as the backbone and conducts the partial decoder using the last three decoders. The details of the decoder are the same as the BAD model, while the only difference is that it is simply trained by a binary cross-entropy loss.

1) Effectiveness of Boundary-Aware Partial Decoder: As we all know, the salient objects in RSIs are usually small,

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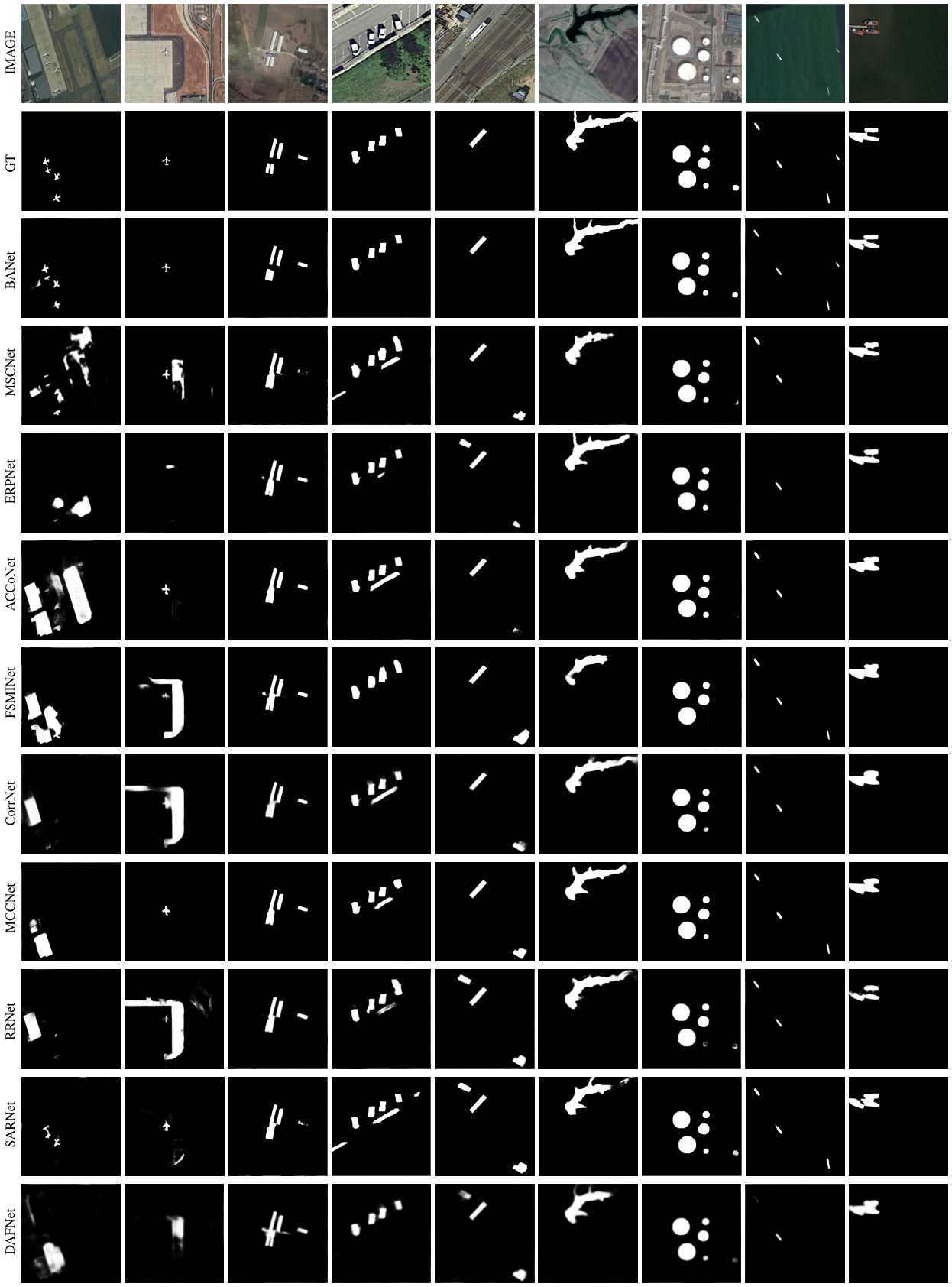

Fig. 10. Visual comparison with the state-of-the-art methods on ORSI-4199. "GT" denotes the ground truth.

raising the pixel-imbalance problem. To validate the effectiveness of the BAD branch, we exploit different types of losses to train the partial decoder. As shown in Table III, the model trained by the combination of the binary cross-entropy loss $\mathcal{L}_{\text{bce}}$ and the dice loss $\mathcal{L}_{\text{dice}}$ outperforms the model trained by the pure $\mathcal{L}_{\text{bce}}$ on all the metrics for all the RSI datasets. This

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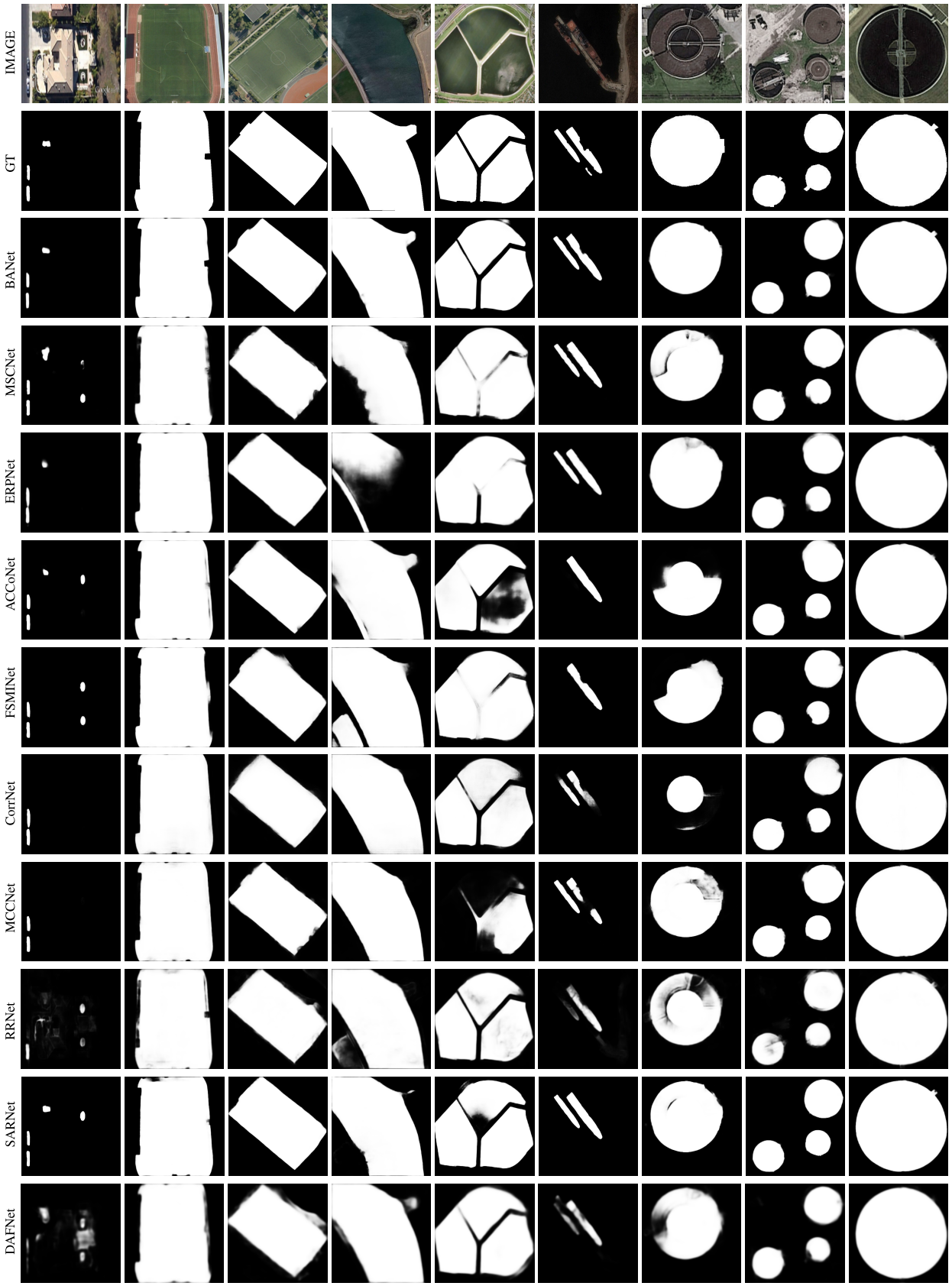

Fig. 11. Visual comparison with the state-of-the-art methods on the constructed RSISOD. "GT" denotes the ground truth.

is contributed to the use of the dice loss for solving the pixel imbalance in RSIs. To further compare the last two rows in Table III, we find that adding the edge-aware loss can further improve the performance of salient objects on all the datasets. This is because the imbalance problem easily appears in the contours of the salient objects. In addition, this submodule

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TABLE II
ABLATION OF KEY MODULES WITHIN THE PROPOSED BANET. BEST RESULTS ARE SHOWN IN BOLD

BAD	BGFE	SAD	ORSSD			EORSSD			RSISOD		
			$F_\beta^{\max}\uparrow$	$S_\alpha\uparrow$	$\mathcal{M}\downarrow$	$F_\beta^{\max}\uparrow$	$S_\alpha\uparrow$	$\mathcal{M}\downarrow$	$F_\beta^{\max}\uparrow$	$S_\alpha\uparrow$	$\mathcal{M}\downarrow$
✓			0.8975	0.9214	0.0129	0.8763	0.9064	0.0107	0.8739	0.8891	0.0264
✓	✓		0.8993	0.9227	0.0118	0.8789	0.9106	0.0102	0.8761	0.8938	0.0231
✓	✓	✓	**0.9205**	**0.9432**	**0.0097**	**0.9055**	**0.9292**	**0.0089**	**0.9073**	**0.9083**	**0.0205**

TABLE III
BAD BRANCH TRAINED USING DIFFERENT LOSSES, INCLUDING $\mathcal{L}_{\text{bce}}$, $\mathcal{L}_{\text{dice}}$, AND $\mathcal{L}_{\text{edge}}$. BEST RESULTS ARE SHOWN IN BOLD

$\mathcal{L}_{bce}$	$\mathcal{L}_{dice}$	$\mathcal{L}_{edge}$	ORSSD			EORSSD			RSISOD		
			$F_\beta^{\max}\uparrow$	$S_\alpha\uparrow$	$\mathcal{M}\downarrow$	$F_\beta^{\max}\uparrow$	$S_\alpha\uparrow$	$\mathcal{M}\downarrow$	$F_\beta^{\max}\uparrow$	$S_\alpha\uparrow$	$\mathcal{M}\downarrow$
✓			0.8930	0.9148	0.0134	0.8691	0.8990	0.0112	0.8656	0.8814	0.0273
✓	✓		0.8961	0.9183	0.0132	0.8725	0.9034	0.0109	0.8694	0.8857	0.0268
✓	✓	✓	**0.8975**	**0.9214**	**0.0129**	**0.8763**	**0.9064**	**0.0107**	**0.8739**	**0.8891**	**0.0264**

TABLE IV
SAD BRANCH TRAINED BY THE TOTAL LOSS COMBINED WITH $\mathcal{L}_{\text{bce}}$ AND $\mathcal{L}_{\text{ssim}}$ IS APPENDED INTO THE BAD BRANCH. BEST RESULTS ARE SHOWN IN BOLD

$\mathcal{L}_{BAD}$	$\mathcal{L}_{bce}$	$\mathcal{L}_{ssim}$	ORSSD			EORSSD			RSISOD		
			$F_\beta^{\max}\uparrow$	$S_\alpha\uparrow$	$\mathcal{M}\downarrow$	$F_\beta^{\max}\uparrow$	$S_\alpha\uparrow$	$\mathcal{M}\downarrow$	$F_\beta^{\max}\uparrow$	$S_\alpha\uparrow$	$\mathcal{M}\downarrow$
✓	✓		0.9136	0.9408	0.0111	0.8985	0.9239	0.0091	0.8970	0.9023	0.0211
✓		✓	0.9158	0.9411	0.0102	0.9023	0.9272	0.0091	0.9029	0.9061	0.0208
✓	✓	✓	**0.9205**	**0.9432**	**0.0097**	**0.9055**	**0.9292**	**0.0089**	**0.9073**	**0.9083**	**0.0205**

takes 15.5 M parameters and 82.52G flops. It costs around 106 frames/s for inference time.

2) Effectiveness of BGFE Module: To verify the effectiveness of the proposed BGFE, we compare the BAD branch with and without the BGFE module. Here, the BAD branch with the BGFE module includes two-part losses, including one $\mathcal{L}_{\text{BAD}}$ for BAD and another $\mathcal{L}_{\text{bce}}$ for the BGFE module to get the final saliency map. As the compared results shown in Table II, adding the BGFE module can further improve the performance of the saliency map on all the metrics. These experimental results also validate the assumption stated in Section III that the high-level features include enough effective information for recovering the encoded features into the final saliency map.

3) Effectiveness of Structure-Aware Partial Decoder: To further validate the effectiveness of the second-stage partial decoder, we cascade the SAD branch after the BGFE module and train it using the structural-similarity loss to make the predicted salient objects have complete structural integrity. As the experimental data recorded in Table II, appending the SAD branch into the BGFE module can further improve the prediction performance of salient objects on all the datasets, which validates the effectiveness of such a two-stage decoder architecture. Moreover, to further verify the effectiveness of using the combination of the binary cross-entropy loss $\mathcal{L}_{\text{bce}}$ and structural-similarity loss $\mathcal{L}_{\text{ssim}}$, we use the variety of losses to optimize the whole model. Table IV demonstrates that the model trained using $\mathcal{L}_{\text{ssim}}$ outperforms the model trained by $\mathcal{L}_{\text{bce}}$. Through further observation, we find that the model trained by the combination of $\mathcal{L}_{\text{ssim}}$ and $\mathcal{L}_{\text{bce}}$ can achieve the best results on all the datasets, validating the effectiveness of our proposed structure-aware loss.

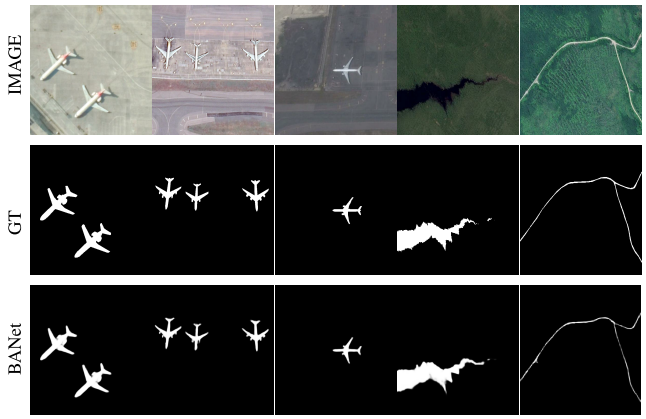

Fig. 12. Visual examples of the RSI saliency maps on ORSI-4199, which are predicted by our proposed BANet, are provided to show the limitation of BANet.

F. Limitations

Last but not least, Fig. 12 provides the visual examples of the limitation on our proposed BANet. These qualitative examples illustrate that the proposed BANet has a limited ability to predict the results with very precise structures when the salient objects are very tiny and their structures are quite complex. However, our method can still generate small salient objects with clear contour, smooth edges, and complete structure. This also verifies that our method can achieve a good prediction of a complicated scenario in RSIs.

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VI. Conclusion

In this article, we propose a BANet with two-stage partial decoders, namely, BANet, for exploring the SOD in optical RSIs. In BANet, we first introduce a partial decoder which is guided by our proposed edge-aware binary dice loss to balance the pixels between the foreground and the background, generating the smooth contours of the salient objects. The resulting features are forwarded into the BGFE module to refine the high-level features conversely. Then, we design another partial decoder which is supervised by our developed structure-aware loss function, recovering the final salient objects with the complete structure. Benefiting from the partial decoder structure, our BANet can fully leverage the valid information and ignore the invalid information to predict high-quality saliency map. Besides, since the existing benchmark datasets are not large-scale enough and cover limited scenes, we also construct a new large-scale challenging dataset with abundant scenarios for SOD in optical RSIs. Extensive experimental results validate the effectiveness and superiority of our proposed BANet method. In the future, we intend to develop a lightweight model with smaller size and will introduce model pruning techniques to remove the redundant parameters from our model, further accelerating the model inference on result prediction.

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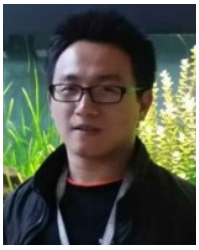

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